DAILY ACTIVE RECALL · MASTER MATRIX & FACTS

ENVIRONMENT & ECOLOGY

पर्यावरण एवं पारिस्थितिकी — उच्च-प्राथमिकता पुनरावलोकन

Environmental Legislation & Acts, Wildlife Projects, Biosphere Reserves, Ramsar Wetlands in UP & UK, International Conventions (UNFCCC, CBD, Ramsar), and IUCN Red List Status.

- **High-Yield Ratta Matrix**
  Concentrated prelims facts, tables & comparisons
- **Examiner Trap Eliminators**
  Confused pairs and high-frequency exam traps
- **15–20 Min Daily Read**
  Formulated for active morning memory drills
- **Exam-Aligned Coverage**
  Curated strictly for UPPSC Prelims, RO/ARO & BEO

UP-PCS & UK-PCS Examination Resource
General Studies Paper 1 · Prelims Fast-Recall Desk

Daily Revision Master Edition · 2026
Nitish Pratap Official Library

Daily Read — High-Yield Revision Facts (Environment & Ecology)

★ UPPCS & UKPCS Prelims Daily Fact Dump · Active Recall Cheat-Sheet

Exam Focus: UPPSC Prelims (Paper 1), UKPSC Prelims (Paper 1), RO/ARO, BEO
Frequency: Daily morning read (15–20 mins)
Question Types Covered: Environmental Acts vs Years, International Conventions & Protocols, 18 Biosphere Reserves & UNESCO MAB list, Wildlife Conservation Projects, Environmental Movements & Pioneers, Research Institutes, Ecological Pyramids & Food Chains, AQI pollutants, Ramsar Sites in UP & UK, Environmental Days, Ecological Concepts & Coining Pioneers, Toxic Pollutants & Occupational Diseases, 4 Biodiversity Hotspots of India.

1. Core Environmental Legislation in India (Exact Years & Articles)

Legislation / Policy Name	Year of Enactment	Constitutional Backing / Key Mandate	Landmark Institutional Outcome
Indian Forest Act	**1927** (earlier 1865, 1878)	Consolidated forest laws; categorized into **Reserved, Protected, and Village Forests**	State Forest Departments
Wildlife (Protection) Act (WPA)	**1972** (9 Sept 1972)	Framework for protecting wild animals, birds, and plants; **6 original schedules restructured into 4 schedules in 2022 Amendment**	National Board for Wildlife (NBWL, chaired by PM); NTCA; Wildlife Crime Control Bureau (WCCB)
Water (Prevention and Control of Pollution) Act	**1974**	Prevention and abatement of water pollution; maintains wholesomeness of water	Established **Central Pollution Control Board (CPCB)** & State PCBs
Forest (Conservation) Act (FCA)	**1980** (25 Oct 1980)	Prohibits diversion of forest land for non-forest purposes without prior central approval	Forest clearance mechanism
Air (Prevention and Control of Pollution) Act	**1981**	Prevention, control, and abatement of air pollution; **Noise pollution included as an air pollutant by 1987 Amendment**	Empowered CPCB and SPCBs to set ambient air standards
Environment (Protection) Act (EPA)	**1986** (23 May 1986)	**Umbrella legislation** enacted under **Article 253** in the wake of the **Bhopal Gas Tragedy (Dec 1984)**	Empowered Union Govt to coordinate state actions, set emission norms; CRZ notifications; EIA rules
Public Liability Insurance Act	**1991**	Mandatory public insurance for providing immediate relief to victims of accidents occurring while handling hazardous substances	Environment Relief Fund
National Environment Tribunal Act	**1995**	Strict liability for damages arising out of any accident while handling hazardous substances	Precursor to NGT
Biological Diversity Act (BDA)	**2002**	Realized commitments under CBD (1992); 3-tier structure for conservation and equitable benefit sharing	**National Biodiversity Authority (NBA, Chennai)**, State Biodiversity Boards (SBB), Biodiversity Management Committees (BMC)
Forest Rights Act (FRA)	**2006**	Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act; vests forest land rights	Gram Sabha is the statutory authority to initiate rights recognition process

Legislation / Policy Name	Year of Enactment	Constitutional Backing / Key Mandate	Landmark Institutional Outcome
National Green Tribunal (NGT) Act	**2010** (18 Oct 2010)	Created dedicated environmental tribunal under **Article 21 (Right to a healthy environment)**; mandated disposal within **6 months**	**Principal Bench: New Delhi**; 4 Regional Benches: **Bhopal, Pune, Kolkata, Chennai**
CAMPA Act	**2016**	Compensatory Afforestation Fund Act; manages funds collected for diversion of forest land (90% to State, 10% to Centre)	National & State Compensatory Afforestation Funds

2. Environmental Movements in India (Pioneers & Locations)

Movement	Year & Place	Key Leaders / Activists	Core Environmental Objective / Strategy
Bishnoi Movement	**1730**, Khejarli village (Jodhpur, Rajasthan)	**Amrita Devi Bishnoi**	King Abhay Singh ordered felling of **Khejri trees** (*Prosopis cineraria*); Amrita Devi and her 3 daughters hugged trees; **363 Bishnois sacrificed their lives**; Amrita Devi Bishnoi Wildlife Protection Award instituted in memory.
Chipko Movement	**1973**, Reni village, Chamoli (**Uttarakhand**)	**Sunderlal Bahuguna**, **Chandi Prasad Bhatt** (Ramon Magsaysay awardee), **Gaura Devi** (*Chipko Mother*)	Protest against commercial logging of ash trees in Alaknanda basin; villagers hugged trees to prevent felling; coined slogan: "*Ecology is permanent economy*"; led to 15-year ban on green felling in UP/UK hills.
Appiko Movement	**1983**, Salkani, Uttara Kannada (Karnataka)	**Pandurang Hegde**	Southern version of Chipko; hugged trees in Western Ghats to save tropical forests from commercial monoculture plantations.
Silent Valley Movement	**1978**, Palakkad (Kerala)	Kerala Sastra Sahitya Parishad (**KSSP**), Romulus Whitaker	Protest against proposed **hydroelectric project on Kunthipuzha River** that would submerge the pristine moist evergreen rainforest habitat of the endangered **Lion-Tailed Macaque**; project scrapped; declared Silent Valley National Park (1984).
Narmada Bachao Andolan (NBA)	**1985**, Narmada Valley (Gujarat, MP, Maha)	**Medha Patkar**, **Baba Amte**	Mass movement against large dams on Narmada (particularly **Sardar Sarovar Dam**); demanded comprehensive rehabilitation, environmental impact assessment, and rights of displaced tribal villagers.
Jungle Bachao Andolan	**1982**, Singhbhum (Bihar / Jharkhand)	Local tribal communities	Tribals agitated against government scheme to replace indigenous, multipurpose **Sal forests** with commercially profitable **Teak (Sagwan)** plantations; known as "Greed Game Political Populism".
Tehri Dam Protest	1980s–1990s, Tehri (**Uttarakhand**)	**Sunderlal Bahuguna**	Agitation against the construction of the mega Tehri Dam on the seismic fault lines of Bhagirathi River; multiple hunger strikes.

3. Environmental Research Institutes & Organizations (Match Goldmine)

Institute / Organization	Location / Headquarters	Established Year / Key Function
Botanical Survey of India (BSI)	**Kolkata** (West Bengal)	**1890**; survey and conservation of plant resources of India
Zoological Survey of India (ZSI)	**Kolkata** (West Bengal)	**1916**; survey, exploration, and taxonomic research on animal diversity

Institute / Organization	Location / Headquarters	Established Year / Key Function
Forest Research Institute (FRI)	**Dehradun (Uttarakhand)**	**1906** (Imperial Forest Research Institute); Greco-Roman architectural marvel
Wildlife Institute of India (WII)	**Dehradun (Uttarakhand)**	**1982**; premier autonomous institution for wildlife research and management
National Environmental Engineering Research Institute (NEERI)	**Nagpur** (Maharashtra)	**1958**; pioneer in environmental monitoring, sewage, industrial pollution solutions (CSIR)
Central Pollution Control Board (CPCB)	**New Delhi**	**1974**; statutory organization under Water Act 1974; sets national air/water norms
National Biodiversity Authority (NBA)	**Chennai** (Tamil Nadu)	**2003**; statutory body under Biological Diversity Act 2002
Sálim Ali Centre for Ornithology & Natural History (SACON)	**Coimbatore** (Tamil Nadu)	Dedicated to conservation of avifauna and wetland ecology
Indian Institute of Forest Management (IIFM)	**Bhopal** (Madhya Pradesh)	**1982**; premier forestry management and planning institute
G.B. Pant National Institute of Himalayan Environment	**Kosi-Katarmal, Almora (Uttarakhand)**	**1988**; apex research body on sustainable development and ecology of Indian Himalayas
Central Arid Zone Research Institute (CAZRI)	**Jodhpur** (Rajasthan)	**1959**; research on desertification and arid land agriculture
National Disaster Management Authority (NDMA)	**New Delhi**	**2005**; apex body under Disaster Management Act 2005; **Chaired by the Prime Minister**

4. Indian Wildlife Conservation Projects (Chronological Ladder)

Project Name	Year Launched	Launch Location / Nodal Agency	Key Objective / Status / Prelims Fact
Project Tiger	**1 April 1973**	**Jim Corbett National Park (Uttarakhand)**	Central scheme; initiated with 9 reserves under Kailash Sankhala ("Tiger Man of India"); statutory backing via NTCA (2006); **58 Tiger Reserves** (Madhav TR, MP, is the 58th). Corbett has among the highest tiger densities.
Project Crocodile	**1975**	Hyderabad / Odisha (with UNDP & FAO aid)	Conservation of 3 crocodilian species: Gharial (*Gavialis gangeticus*), Mugger crocodile, and Saltwater crocodile (Bhitarkanika, Odisha)
Project Hangul	**1970**	Dachigam National Park (J&K)	Conservation of Kashmir Stag (*Cervus elaphus hanglu* - critically endangered) with IUCN/WWF aid
Project Rhino (IRV 2020)	**1987 / 2005**	Assam (Kaziranga, Manas, Orang, Pobitora)	Indian Rhino Vision 2020 aimed to attain a wild population of at least 3,000 Greater One-Horned Rhinos in Assam
Project Elephant	**February 1992**	Jharkhand (Singhbhum)	Centrally sponsored; protects elephant habitats and corridors; addresses human-elephant conflict; **Mike sites** (Monitoring of Illegal Killing of Elephants); campaign: *Haathi Mere Saathi*
Project Sea Turtle	**1999**	Odisha coast (Gahirmatha / Rushikulya)	Conservation of **Olive Ridley turtles** (*Lepidochelys olivacea*) during synchronized mass nesting (**Arribada**) with MoEFCC & UNDP
Project Snow Leopard	**2009**	High-altitude Himalayan states (Hemis, Gangotri, Nanda Devi)	Conservation of high-altitude ecosystem and the elusive Ghost of the Mountains (*Panthera uncia*)

Project Name	Year Launched	Launch Location / Nodal Agency	Key Objective / Status / Prelims Fact
Project Dolphin	**2020**	Ministry of Environment, Forest & Climate Change	Gangetic Dolphin (*Platanista gangetica*) declared **National Aquatic Animal on 5 October 2009** (5 Oct celebrated as **National Dolphin Day**); blind; uses echolocation; Vikramshila Sanctuary (Bihar)
Project Cheetah	**September 2022**	**Kuno National Park (Madhya Pradesh)**	Intercontinental translocation of African cheetahs from Namibia (8) and South Africa (12); Asiatic Cheetah declared extinct in India in **1952**

5. Ramsar Wetlands in Uttar Pradesh & Uttarakhand

- **Uttarakhand**:
- **Asan Conservation Reserve** (Dehradun; confluence of Asan and Yamuna; **first Ramsar of Uttarakhand, Oct 2020**).
- **Jhilmil Jheel Conservation Reserve** (Haridwar, **2005**) is the state's swamp-deer / barasingha wetland — **not** the Ramsar first (that is Asan).
- **Uttar Pradesh (12 Ramsar sites)** — do not stop at the older list of 10:
- **Bakhira Wildlife Sanctuary** (Sant Kabir Nagar).
- **Haiderpur Wetland** (Muzaffarnagar–Bijnor; Solani / Hastinapur WLS).
- **Nawabganj Bird Sanctuary** (*Chandra Shekhar Azad*, Unnao).
- **Parvati Arga Bird Sanctuary** (Gonda).
- **Saman Bird Sanctuary** (Mainpuri).
- **Samaspur Bird Sanctuary** (Rae Bareli).
- **Sandi Bird Sanctuary** (Hardoi).
- **Sarsai Nawar Jheel** (Etawah; Sarus Crane).
- **Sur Sarovar (Keetham Lake)** (Agra).
- **Upper Ganga River** (Brijghat to Narora).
- **Patna Bird Sanctuary** (Etah).
- **Shekha Jheel** (Aligarh).

> **Traps:** Sultanpur NP = **Haryana**. Rudrasagar = **Tripura**. Samaspur and Sarsai Nawar **are** in UP.

5A. Protected areas of UP & UK — Declared / still in force

PA	State	From (year)	High-yield
Hailey → Ramganga → Jim Corbett NP / TR	**Uttarakhand**	NP **1936** (Hailey); Project Tiger launch site **1973**	India's first National Park. Nainital–Pauri belt.
Rajaji NP / TR	**Uttarakhand**	NP **1983**; tiger reserve **2015**	Elephant corridor; Haridwar–Dehradun–Pauri
Nanda Devi NP	**Uttarakhand**	**1982** (BR **1988**; UNESCO mixed serial with VoF)	Chamoli; Nanda Devi peak
Valley of Flowers NP	**Uttarakhand**	**1982** (UNESCO **1988**, expanded **2005**)	Smallest UK NP; Chamoli
Gangotri NP	**Uttarakhand**	**1989**	Largest UK NP (~2390 sq km); Uttarkashi
Govind NP / WLS	**Uttarakhand**	WLS **1955**; NP later	Snow leopard belt; Uttarkashi
Dudhwa NP / TR	**Uttar Pradesh**	NP **1977**; TR **1987**	Lakhimpur Kheri Terai; rhino from **1984**; barasingha
Pilibhit TR	**Uttar Pradesh**	**2014**	Terai; tiger + woodland swamp deer

PA	State	From (year)	High-yield
Ranipur TR	**Uttar Pradesh**	**2022/2023**	Chitrakoot–Banda (Vindhyan); not Terai
Hastinapur WLS	**Uttar Pradesh**	**1986**	**Largest UP sanctuary**; Meerut–Bijnor belt; Haiderpur Ramsar inside
Katarniaghat WLS	**Uttar Pradesh**	**1975**	Terai Arc toward Nepal; gharial / dolphin

UP has **no** biosphere reserve and **no** marine PA. UK biosphere = **Nanda Devi (1988; UNESCO MAB 2004)**.

6. National Air Quality Index (AQI)

Launched in **October 2014** under Swachh Bharat Abhiyan by CPCB: - **Six Categories**: 1. *Good* (0–50) 2. *Satisfactory* (51–100) 3. *Moderately Polluted* (101–200) 4. *Poor* (201–300) 5. *Very Poor* (301–400) 6. *Severe* (401–500). - **Eight Monitored Pollutants (Frequently Tested in PCS!)**: 1. **Particulate Matter 10 (PM10)** 2. **Particulate Matter 2.5 (PM2.5)** 3. **Nitrogen Dioxide (NO2)** 4. **Sulphur Dioxide (SO2)** 5. **Carbon Monoxide (CO)** 6. **Ozone (O3)** 7. **Ammonia (NH3)** 8. **Lead (Pb)**. *(Notice: Carbon Dioxide (CO2) is NOT part of the AQI 8 pollutants!)*

7. The 18 Biosphere Reserves of India & 12 UNESCO MAB Reserves

No.	Biosphere Reserve	Year Set Up	State / Region	Part of UNESCO MAB (WNBR)?
1	**Nilgiri**	**1986**	Tamil Nadu, Kerala, Karnataka	**Yes (2000 — First in India)**
2	**Nanda Devi**	**1988**	**Uttarakhand** (Chamoli, Pithoragarh, Bageshwar)	**Yes (2004)**
3	**Nokrek**	**1988**	Meghalaya (Garo Hills)	**Yes (2009)**
4	**Great Nicobar**	**1989**	Andaman and Nicobar Islands	**Yes (2013)**
5	**Gulf of Mannar**	**1989**	Tamil Nadu (marine)	**Yes (2001)**
6	**Manas**	**1989**	Assam	No
7	**Sundarbans**	**1989**	West Bengal	**Yes (2001)**
8	**Similipal**	**1994**	Odisha (Mayurbhanj)	**Yes (2009)**
9	**Dibru-Saikhowa**	**1997**	Assam (*Smallest Biosphere Reserve*)	No
10	**Dehang-Debang**	**1998**	Arunachal Pradesh	No
11	**Pachmarhi**	**1999**	Madhya Pradesh (Satpura Range)	**Yes (2009)**
12	**Khangchendzonga**	**2000**	Sikkim	**Yes (2018)**
13	**Agasthyamalai**	**2001**	Kerala and Tamil Nadu	**Yes (2016)**
14	**Achanakmar-Amarkantak**	**2005**	Madhya Pradesh and Chhattisgarh	**Yes (2012)**
15	**Great Rann of Kutch**	**2008**	Gujarat (*Largest Biosphere Reserve*)	No
16	**Cold Desert**	**2009**	Himachal Pradesh (Pin Valley, Kibber, Chandratal)	No
17	**Seshachalam Hills**	**2010**	Andhra Pradesh (Tirupati)	No
18	**Panna**	**2011**	Madhya Pradesh	**Yes (2020)**

8. International Environmental Conventions & Protocols

Year	Convention / Agreement	Place / City	Primary Objective / Regulatory Scope
1971 (2 Feb)	**Ramsar Convention on Wetlands**	Ramsar (Iran)	Conservation and wise use of wetlands; entered into force 1975; **World Wetlands Day (2 Feb)**; India joined **1 Feb 1982**; Keoladeo & Chilika were first sites; **Montreux Record** (Keoladeo NP & Loktak Lake)
1972 (5–16 Jun)	**Stockholm Conference (UNCHE)**	Stockholm (Sweden)	UN Conference on the Human Environment; **Created UNEP (Nairobi, Kenya)**; celebrated as **World Environment Day (5 June)**
1973 (3 Mar)	**CITES (Washington Convention)**	Washington, D.C. (USA)	Convention on International Trade in Endangered Species of Wild Fauna and Flora; Appendix I, II, III; **World Wildlife Day (3 March)**
1979	**Bonn Convention (CMS)**	Bonn (Germany)	Convention on the Conservation of Migratory Species of Wild Animals
1982	**UNCLOS**	Montego Bay (Jamaica)	Law of the Sea: Territorial waters (12 nm), Contiguous zone (24 nm), Exclusive Economic Zone (200 nm)
1985	**Vienna Convention**	Vienna (Austria)	Framework convention for the protection of the stratospheric ozone layer
1987 (16 Sep)	**Montreal Protocol**	Montreal (Canada)	**Legally binding phase-out of Ozone Depleting Substances (ODS / CFCs)**; entered into force 1 Jan 1989; **World Ozone Day (16 Sept)**; **Kigali Amendment (2016)** phase out HFCs
1987	**Brundtland Report**	UN World Commission	Report titled ***Our Common Future***; formally defined **Sustainable Development**
1989	**Basel Convention**	Basel (Switzerland)	Control of transboundary movements of **Hazardous Wastes and their disposal**; entered into force 1992
1992 (3–14 Jun)	**Rio Earth Summit (UNCED)**	Rio de Janeiro (Brazil)	Landmark summit producing 3 Rio Conventions: 1. **UNFCCC** (Climate Change) 2. **CBD** (Biological Diversity) 3. **UNCCD** (Combating Desertification) Also produced: **Agenda 21**, Rio Declaration, and Forest Principles
1997 (11 Dec)	**Kyoto Protocol**	Kyoto (Japan)	UNFCCC COP-3; **Legally binding greenhouse gas emission reduction targets for Annex-I developed countries**; 6 greenhouse gases (CO2, CH4, N2O, HFCs, PFCs, SF6); Clean Development Mechanism (CDM); entered into force 2005; expired 2020
1998	**Rotterdam Convention**	Rotterdam (Netherlands)	Prior Informed Consent (PIC) procedure for certain hazardous chemicals and pesticides in international trade
2000	**Cartagena Protocol on Biosafety**	Montreal / Cartagena	Protocol to CBD governing the safe transfer, handling, and use of **Living Modified Organisms (LMOs)**
2001	**Stockholm Convention on POPs**	Stockholm (Sweden)	Elimination or restriction of **Persistent Organic Pollutants (POPs)** (original "Dirty Dozen" including DDT, Dioxins, PCBs)
2010	**Nagoya Protocol**	Nagoya (Japan)	Supplementary agreement to CBD on **Access to Genetic Resources and the Fair and Equitable Sharing of Benefits (ABS)**
2013	**Minamata Convention**	Kumamoto (Japan)	Global treaty to protect human health and the environment from adverse effects of **Mercury (Hg)**
2015 (12 Dec)	**Paris Agreement**	Paris (France)	UNFCCC COP-21; keep global temperature rise **well below 2°C** above pre-industrial levels, pursue efforts to limit rise to **1.5°C**; Nationally Determined Contributions (NDCs); entered into force **4 Nov 2016**

Year	Convention / Agreement	Place / City	Primary Objective / Regulatory Scope
2015	**Sustainable Development Goals (SDGs)**	UN General Assembly	**17 Goals, 169 Targets (2015–2030)**; SDG 1: No Poverty, SDG 2: Zero Hunger, SDG 3: Good Health, SDG 4: Quality Education, SDG 5: Gender Equality, SDG 6: Clean Water & Sanitation, SDG 13: Climate Action, SDG 14: Life Below Water, SDG 15: Life on Land
2022	**Kunming-Montreal GBF**	Montreal (CBD COP-15)	Global Biodiversity Framework; includes landmark **"30x30 target"** (protect at least 30% of planet's land and ocean by 2030)

9. Fundamental Ecological Principles & Laws

A. Lindeman's 10% Energy Law (1942)

- Formulated by **Raymond Lindeman**.
- Only **approximately 10% of the energy** entering any trophic level is transferred and organized into organic matter at the next higher trophic level.
- 90% of energy is lost as metabolic heat, respiration, and catabolism.
- Explains why food chains rarely exceed 4 to 5 trophic levels.

B. Ecological Pyramids (Charles Elton, 1927)

1. **Pyramid of Energy**:
2. **ALWAYS UPRIGHT, WITHOUT EXCEPTION!**
3. Energy decreases progressively at each successive trophic level according to thermodynamics.
4. **Pyramid of Biomass**:
5. **Terrestrial Ecosystems (Grassland, Forest)**: **UPRIGHT** (Producer biomass > Herbivore > Carnivore).
6. **Aquatic Ecosystems (Pond, Lake, Ocean)**: **INVERTED** (Phytoplankton standing crop biomass is much smaller than zooplankton and predatory fish).
7. **Pyramid of Numbers**:
8. **Grassland & Cropland**: **UPRIGHT** (Millions of grasses support fewer herbivores, and even fewer carnivores).
9. **Tree / Parasitic Ecosystem**: **INVERTED** (A single large tree supports hundreds of birds, which host thousands of parasites and hyper-parasites).
10. **Forest Ecosystem**: **Spindle-shaped** (Few large trees support many herbivores, which are eaten by fewer carnivores).

C. Bioaccumulation vs Biomagnification

- **Bioaccumulation**: Increase in the concentration of a chemical pollutant in an **individual organism over time** (rate of intake > rate of excretion).
- **Biomagnification**: Progressive increase in the concentration of persistent, non-biodegradable, fat-soluble toxins (e.g., **DDT, Mercury, PCBs**) at **successively higher trophic levels** of a food chain.
- *Classic example:* Phytoplankton (0.00005 ppm) → Zooplankton → Small Fish → Fish-eating birds / Falcons (25 ppm causing egg-shell thinning and population collapse).

D. Greenhouse Gases: Global Warming Potential (GWP, 100-Year Horizon)

Ranked from highest warming impact per molecule to lowest (relative to CO2 = 1):

SF6 (23,500) > PFCs (7,000–9,000) > HFCs (1,400–3,000) > N2O (273) > CH4 (28) > CO2 (1)

- **Water Vapour**: Most abundant natural greenhouse gas contributing greatest share to natural greenhouse effect (~60%), but has short atmospheric residence time.
- **Carbon Dioxide (CO2)**: Primary anthropogenic driver of global climate change due to massive volume and century-long residence time.
- **Methane (CH4)**: Major sources: Paddy/rice fields, ruminant livestock enteric fermentation, landfills, wetlands, coal mining.

10. Important Environmental Days (UPPCS Direct Match)

Date	Environmental Observance	Connected Milestone / Significance
2 February	**World Wetlands Day**	Signing of the Ramsar Convention on Wetlands (1971)
3 March	**World Wildlife Day**	Adoption of CITES (Washington Convention, 1973)
21 March	**International Day of Forests**	Promotes awareness of the importance of all types of forests
22 March	**World Water Day**	Focuses on the importance of freshwater
23 March	**World Meteorological Day**	Establishment of WMO (1950)
22 April	**Earth Day**	First held in 1970; global environmental movement
22 May	**International Day for Biological Diversity**	Adoption of the agreed text of the CBD (1992)
5 June	**World Environment Day**	Opening day of the historic Stockholm Conference (1972)
8 June	**World Oceans Day**	Concept originally proposed at Rio Earth Summit 1992
29 July	**International Tiger Day**	Instituted at the Saint Petersburg Tiger Summit (2010)
12 August	**World Elephant Day**	Dedicated to the preservation and protection of elephants
16 September	**World Ozone Day**	Signing of the Montreal Protocol (1987)
5 October	**National Dolphin Day (India)**	Declaration of Gangetic Dolphin as National Aquatic Animal (2009)
16 October	**World Food Day**	Founding of the FAO (1945)
5 December	**World Soil Day**	Promotes importance of healthy soil and sustainable soil management

11. Foundational Ecological Terms & Coining Scientists / Pioneers

Term / Ecological Concept	Coined By / Pioneer	Year	Core Definition / Exam Relevance
Ecology (*Oekologie*)	**Ernst Haeckel**	**1866**	Coined from Greek *oikos* (house/dwelling) + *logos* (study). Study of interactions between living organisms and their physical environment.
Ecosystem	**Arthur Tansley (A.G. Tansley)**	**1935**	The basic functional ecological unit consisting of biotic community together with its abiotic environment interacting as a system.
Biodiversity (Biological Diversity)	**Walter G. Rosen** (coined); popularized by **Edward O. Wilson (E.O. Wilson)**	**1985 / 1988**	Coined as contraction of "biological diversity". E.O. Wilson published landmark book *Biodiversity* (1988); known as "Father of Biodiversity".
Biodiversity Hotspots	**Norman Myers**	**1988**	Biogeographic regions characterized by exceptional levels of plant endemism and high levels of habitat loss (≥70% loss). Adopted by Conservation International.
Ecological Niche	**Joseph Grinnell** (habitat niche); refined by **Charles Elton** & **G.E. Hutchinson**	**1917 / 1927 / 1957**	The specific functional role and physical space occupied by an organism within its ecosystem (*address + profession of species*). Hutchinson introduced "fundamental vs realized niche".
Deep Ecology	**Arne Naess**	**1973**	Radical environmental philosophy rejecting anthropocentrism; advocates that all living beings have intrinsic moral value regardless of their utility to humans.

Term / Ecological Concept	Coined By / Pioneer	Year	Core Definition / Exam Relevance
Ecological Footprint	**William Rees & Mathis Wackernagel**	**1992**	Measure of human demand on Earth's ecosystems; the area of biologically productive land and water required to produce goods consumed and absorb waste.
Gaia Hypothesis	**James Lovelock & Lynn Margulis**	**1972**	Proposes that living organisms interact with their inorganic surroundings on Earth to form a synergistic and self-regulating complex system that helps maintain conditions for life.
Acid Rain	**Robert Angus Smith**	**1872**	Discovered connection between atmospheric pollution (SO2 & NOx from coal burning) and acid precipitation in Manchester, UK. Normal rain pH ~5.6; acid rain pH < 5.6.
Smog (Smoke + Fog)	**H.A. Des Voeux**	**1905**	Coined at the Public Health Congress in London to describe the deadly mixture of coal smoke and dense winter fog (**Classical / London Smog — reducing in nature, contains SO2**).
Photochemical Smog (Los Angeles Smog)	**Arie Haagen-Smit**	**1950s**	Sunlight-induced photochemical reaction between Nitrogen Oxides (NOx) and Volatile Organic Compounds (VOCs). **Oxidizing in nature; major components: Ozone (O3), PAN (Peroxyacetyl nitrate), and Aldehydes**.
Discovery of Antarctic Ozone Hole	**Joe Farman, Brian Gardiner, Jonathan Shanklin**	**May 1985**	British Antarctic Survey scientists discovered drastic springtime thinning of the stratospheric ozone layer over Halley Bay, Antarctica, caused by Chlorofluorocarbons (CFCs).
Biocoenosis	**Karl Möbius**	**1877**	An interacting community of different living species occupying a common biotope/habitat.
Father of Indian Ecology	**Prof. Ramdeo Misra**	1950s–1970s	Established pioneering ecological research and education in India at Banaras Hindu University (BHU), Varanasi.
Ecotone & Edge Effect	**Frederic Clements**	1905	**Ecotone**: Transition zone between two distinct biomes/communities (e.g., mangrove between marine & terrestrial; marshland between aquatic & dry land). **Edge Effect**: Phenomenon where species richness and population density are significantly greater in the ecotone than in either adjoining community (e.g., birds in forest edge).

12. Major Environmental Pollutants, Toxic Heavy Metals & Occupational Diseases

Toxic Pollutant / Metal	Source / Environmental Medium	Characteristic Disease / Syndrome	Clinical Pathology / Prelims Highlights
Methylmercury (Hg)	Industrial effluent discharged into coastal waters; biomagnified in fish	**Minamata Disease**	First discovered in **Minamata Bay (Kumamoto, Japan, 1956)**; caused by Chisso Corp chemical plant; severe neurological syndrome, ataxia, loss of motor control, sensory impairment, convulsions, congenital deformities.
Cadmium (Cd)	Mining runoff, zinc smelting, contaminated irrigation water for paddy fields	**Itai-Itai Disease ("Ouch-Ouch Disease")**	First discovered in **Toyama Prefecture (Japan, 1912)**; severe osteomalacia (extreme bone softening and painful skeletal fractures) and kidney failure; *Itai-Itai* means "it hurts, it hurts" in Japanese.

Toxic Pollutant / Metal	Source / Environmental Medium	Characteristic Disease / Syndrome	Clinical Pathology / Prelims Highlights
Nitrate (NO3-)	Runoff from nitrogenous chemical fertilizers in groundwater drinking water	**Blue Baby Syndrome (Methemoglobinemia)**	Nitrate converts to nitrite in infant gut; binds with infant hemoglobin to form **methemoglobin**, which cannot release oxygen to tissues; skin turns blue due to asphyxiation; fatal if untreated.
Arsenic (As)	Natural geochemical leaching into shallow groundwater aquifers	**Blackfoot Disease, Hyperkeratosis & Arsenicosis**	Severe peripheral vascular gangrene ("black foot"), skin hyper-pigmentation, keratosis, and bladder/skin cancer. Rampant in **Ganga-Brahmaputra alluvial plains: West Bengal, UP, Bihar, and Bangladesh**. Permissible limit: **0.01 mg/L** (WHO).
Fluoride (F-)	Over-extracted groundwater leaching from fluorapatite minerals	**Dental & Skeletal Fluorosis**	Permissible limit: **1.0 to 1.5 mg/L**. Excess causes mottling of teeth, calcification of ligaments, severe crippling skeletal deformity, and **Knock-Knee Syndrome (Genu Valgum)**. Widespread in Rajasthan, Telangana, UP.
Silica (SiO2 dust)	Inhalation of crystalline silica dust in stone quarries, agate cutting, sandblasting	**Silicosis**	Incurable occupational fibrotic lung disease causing permanent pulmonary scarring and progressive respiratory failure.
Asbestos Fibers	Asbestos cement manufacturing, building demolition, brake lining manufacture	**Asbestosis & Mesothelioma**	Asbestos fibers penetrate deep into lungs causing pulmonary fibrosis; strongly carcinogenic, inducing **malignant mesothelioma** (cancer of pleural lining).
Coal Dust	Underground coal mining and extraction	**Pneumoconiosis (Black Lung Disease / Anthracosis)**	Chronic inflammation and irreversible progressive massive fibrosis of lungs due to accumulated inhaled coal mine dust.
Cotton Dust	Cotton mills, textile spinning, and ginning factories	**Byssinosis ("Brown Lung Disease" / Monday Fever)**	Occupational asthma and chest tightness occurring on the first workday after a weekend break (*Monday disease*).
Sugarcane Bagasse	Inhalation of dry bagasse fibers during sugarcane processing & paper milling	**Bagassosis**	Hypersensitivity pneumonitis triggered by thermoactinomycetes in moldy bagasse.
Lead (Pb)	Lead-acid batteries, leaded gasoline (historic), paints, smelters, e-waste	**Plumbism / Saturnism**	Neurotoxic; causes severe cognitive and developmental impairment in children, anemia (inhibits heme synthesis), kidney damage, and characteristic **Burtonian blue line along gums**.
E-Waste Hazardous Elements	Discarded computers, mobile phones, printed circuit boards (PCBs)	Multi-organ toxicity	Lead (solder & CRTs), Cadmium (rechargeable Ni-Cd batteries & chip resistors), Mercury (switches & flat screens), Brominated Flame Retardants (BFRs - endocrine disruption).

13. Biodiversity Hotspots of India & Global Criteria

A. Global Criteria for Biodiversity Hotspot (Norman Myers, 1988)

To qualify as an international biodiversity hotspot, a biogeographic region must fulfill **TWO STRICT QUANTITATIVE CRITERIA**: 1. **High Endemism**: It must contain at least **1,500 species of vascular plants (>0.5% of the world's total) as endemics** (species found nowhere else on Earth). 2. **High Degree of Threat**: It must have lost at least **70% of its original**

primary native vegetation (i.e., has 30% or less of its original natural habitat remaining). - Globally, there are **36 recognized Biodiversity Hotspots** covering only 2.5% of Earth's land surface, but supporting over 50% of world's endemic plant species and 43% of endemic bird, mammal, reptile, and amphibian species.

B. The 4 Biodiversity Hotspots Covering India

Hotspot Name	Geographical Extent in India	Adjoining International Regions	Landmark Endemic Species / Key Features
1. Himalaya Hotspot	Covers the **entire Indian Himalayan region**: Jammu & Kashmir, Ladakh, Himachal Pradesh, **Uttarakhand**, Sikkim, and Arunachal Pradesh.	Pakistan, Nepal, Bhutan, Tibet / Southern China	Snow Leopard, Himalayan Musk Deer, Himalayan Tahr, Blue Sheep (*Bharal*), Red Panda; vast alpine and temperate rhododendron-oak ecosystems.
2. Indo-Burma Hotspot	Covers all of **North-East India south of the Brahmaputra River** (Assam plains/hills, Meghalaya, Nagaland, Manipur, Mizoram, Tripura) and the **Andaman Islands**.	Myanmar, Thailand, Laos, Vietnam, Cambodia, and Southern China	Hoolock Gibbon (India's only ape), Golden Langur, Stump-tailed Macaque; extraordinary diversity of orchids, ginger species, and freshwater turtles.
3. Western Ghats & Sri Lanka Hotspot	Covers the continuous mountain chain along the western coast across **Gujarat, Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu**.	Island nation of Sri Lanka	**Lion-Tailed Macaque** (Silent Valley), **Nilgiri Tahr** (Eravikulam), Nilgiri Langur, Malabar Gliding Frog; exceptional concentration of endemic amphibians (~80% endemic), reptiles, and flowering plants.
4. Sundaland Hotspot	Covers the **Nicobar Islands** of India (Great Nicobar, Little Nicobar).	Indonesia (Sumatra, Java, Borneo), Malaysia, Singapore, Brunei, and Southern Thailand	Nicobar Megapode (mound-building bird), Nicobar Tree Shrew, Saltwater Crocodile; rich coastal coral reefs and dense tropical rainforests.

Crucial Exam Trap: Andaman Islands fall under **Indo-Burma Hotspot**, while Nicobar Islands fall under **Sundaland Hotspot**!